DAY 7 (2)

Network Topologies Theory (1 hour): Types of network topologies: Bus, Star, Ring, Mesh. Practical (1 hour): Simulate a star topology using Cisco Packet Tracer. Task: Draw a diagram of topologies with examples.

Q1. Draw a Diagram of a Bus Topology and Label All Key Components. Also Write One Real-World Example.

Bus Topology Diagram

```
[Terminator]----[PC1]----[PC2]----[PC3]----[PC4]----[Terminator]
                 |
            Backbone Cable
```

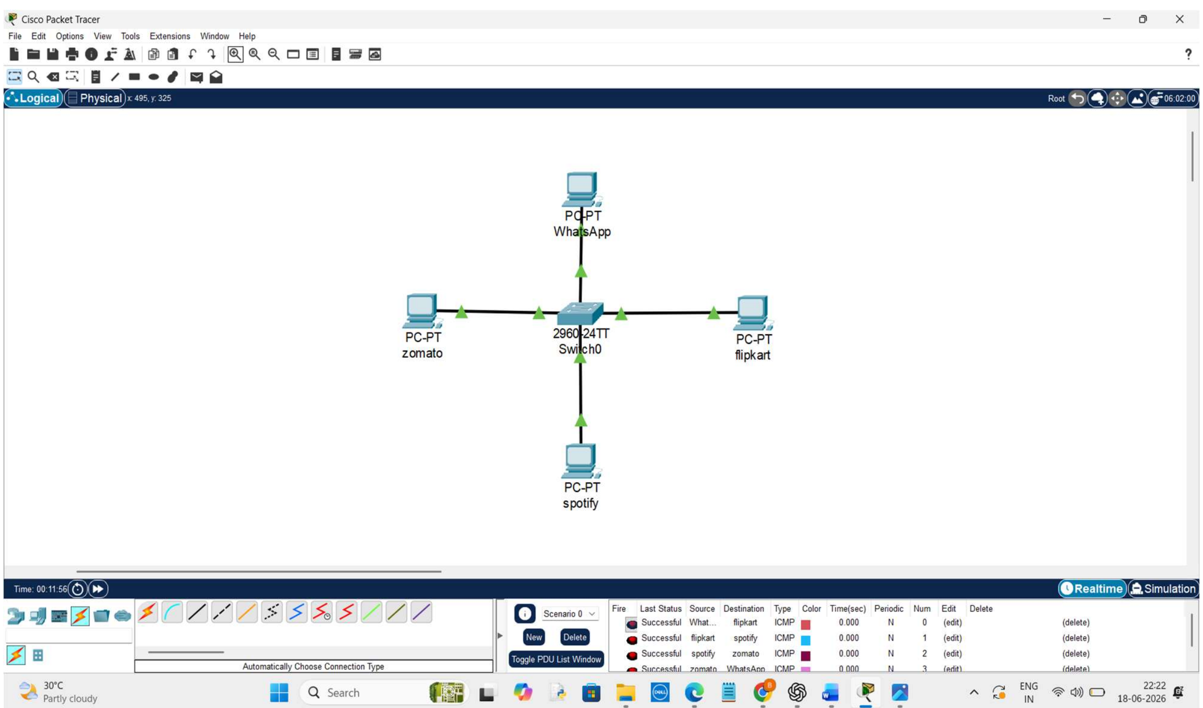
Components

1. **Backbone Cable** – Main communication cable connecting all devices.
2. **PC1, PC2, PC3, PC4** – Network devices connected to the bus.
3. **Terminators** – Devices placed at both ends of the cable to prevent signal reflection.

Real-World Example

A bus topology was commonly used in small office networks and school computer labs using coaxial cables. It was popular in early Ethernet networks because it required less cabling and was inexpensive.

Q2. Create a Simple Star Topology in Cisco Packet Tracer



Q3. Draw a Ring Topology Diagram and Explain Data Flow.

Ring Topology Diagram

```
    [PC1]
   ↗   ↘
[PC4]    [PC2]
   ↖   ↙
    [PC3]
```

Data Flow Direction →

Explanation

In a ring topology, each device is connected to exactly two neighbouring devices, forming a circular path. Data travels from one device to the next in a single direction until it reaches the intended destination. Every device act as a repeater and forwards data to the next device in the ring.

Q4. Compare Mesh and Star Topologies

Comparison Table:

Feature	**Mesh Topology**	**Star Topology**
Structure	Every device is connected to every other device.	All devices connect to a central switch or hub.
Reliability	Very high reliability because multiple paths exist.	Moderate reliability because all communication depends on the central device.
Cost	Expensive due to large number of cables and ports.	Less expensive and easier to install.
Maintenance	Difficult to manage and troubleshoot.	Easy to manage and troubleshoot.

Advantages and Disadvantages:

Topology	**Advantages**	**Disadvantages**
Mesh	1. Highly reliable. 2. No single point of failure.	1. Expensive. 2. Complex installation.
Star	1. Easy to install and manage. 2. Failure of one PC does not affect others.	1. Switch failure affects the whole network. 2. Requires more cable than bus topology.